Homework 9

EE 999 — Introduction to Control Systems

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Due Wednesday, November 18, 2026

Problem 1 — System Identification with `invfreqs()`

Run `HW09_InvfreqsExample.m`. *It fits two transfer functions to simulated frequency-response data: 2 poles, 0 zeros; and 3 poles, 1 zero.*

Problem 1.A

Considering the frequency response plots, which transfer function better fits the data? (The residual appears in the plot title.)

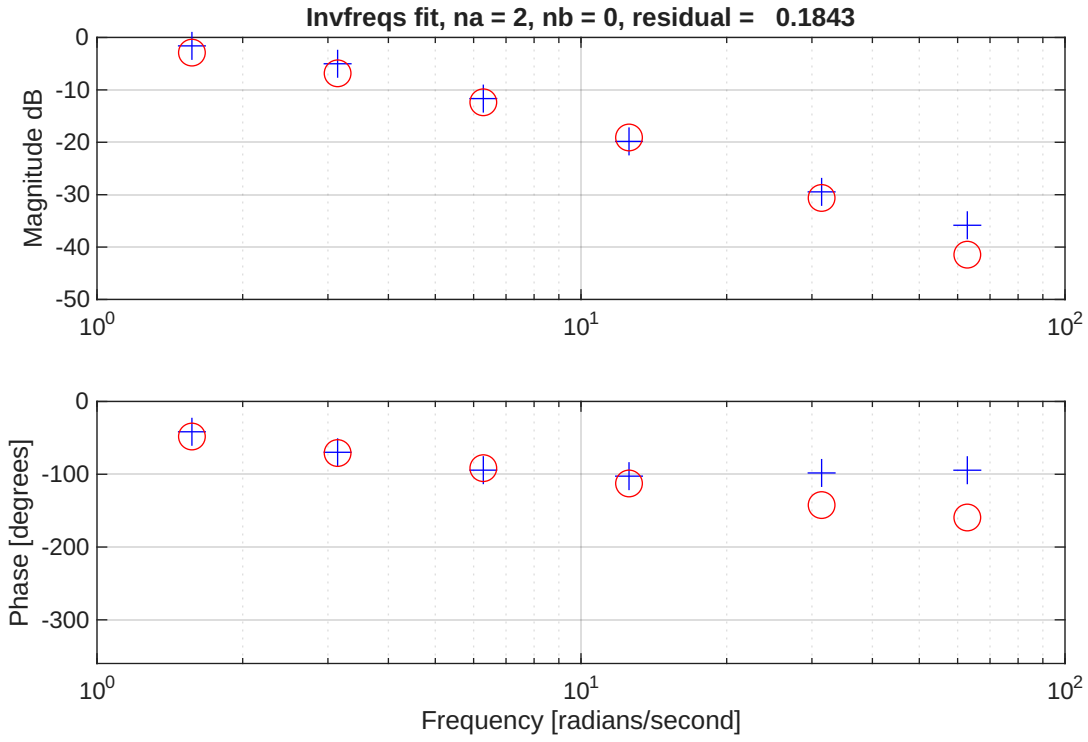


Figure 1: Frequency response data (+) and the 2-pole, 0-zero fit (o).

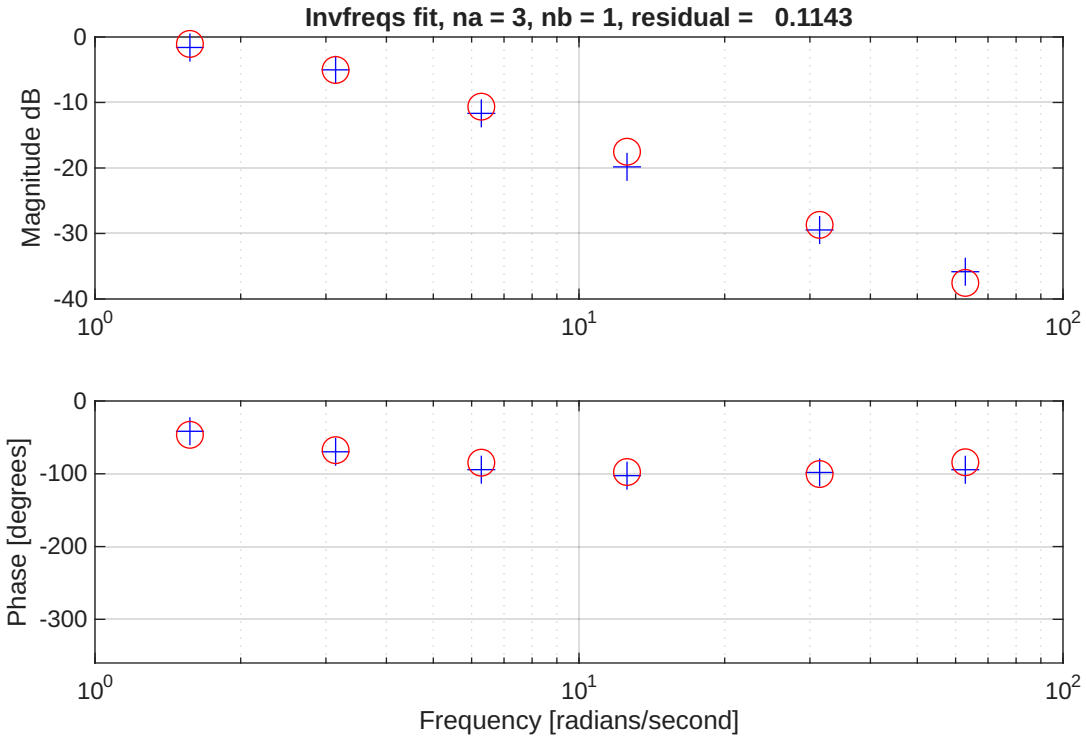


Figure 2: Frequency response data (+) and the 3-pole, 1-zero fit (o).

The transfer function with 3 poles and 1 zero provides a better fit.

Problem 1.B

Considering the pole-zero constellations, is each transfer function stable?

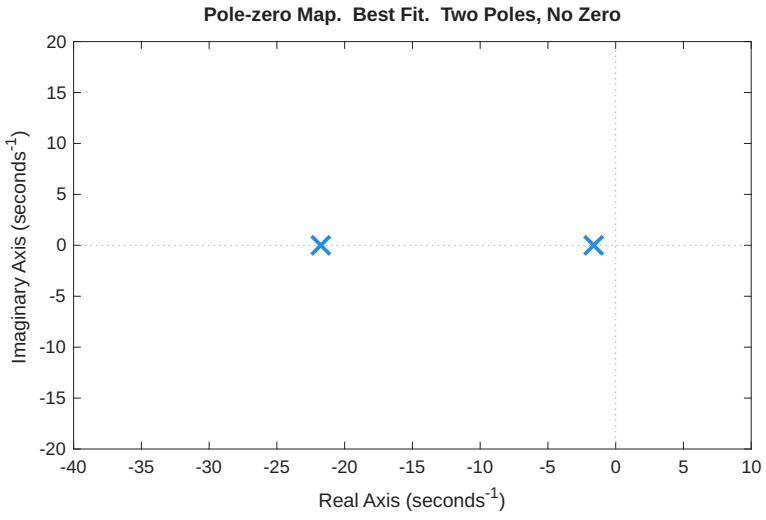


Figure 3: Pole-zero map of the 2-pole, 0-zero fit.

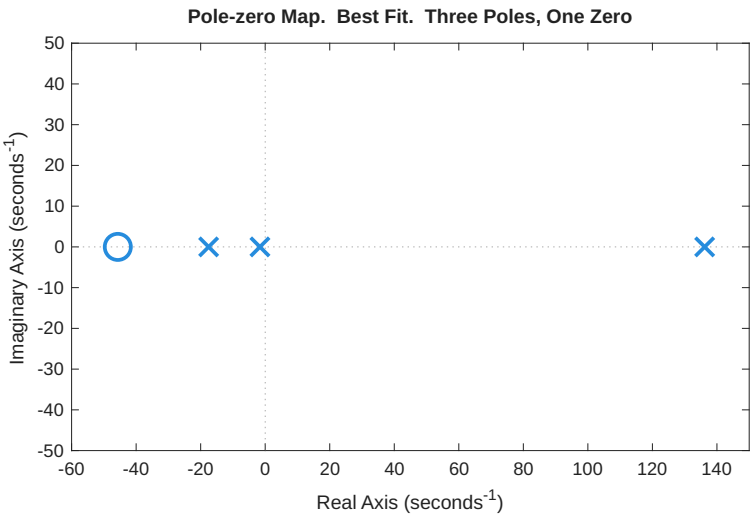


Figure 4: Pole-zero map of the 3-pole, 1-zero fit.

The transfer function with two poles and no zeroes is stable, while the transfer function with three poles and one zero is unstable.

Problem 1.C

Modify the data in `HW09_InvfreqsExample.m` *to be the following. This is the complex gain versus frequency for a system with transfer function* $G_{uy}(s) = B_b(s)/A_b(s)$.

Freq [rad/s]	Mag	Phase [deg]
0.3142	0.4180	-6.0062
0.6283	0.4237	-12.0334
1.2566	0.4426	-24.2801
3.1416	0.5135	-66.4484
6.2832	0.2910	-131.1225
12.5664	0.0754	-161.5733
31.4159	0.0120	-173.4103
62.8319	0.0030	-176.7621

Problem 1.C.1

Determine $G_{uy}(s)$ *for the models (indicate the transfer function):*

Model 1: 2 poles, 0 zeros

write the answer here

Model 2: 3 poles, 1 zero

write the answer here

Problem 1.C.2

Determine the better model based on these criteria:

Is the model physically realistic?

$G_{uy}(s)$ *is known to be open-loop stable — does the model have unstable poles? Does it have poles or zeros substantially faster than the fastest data point,* $\omega_{fastest} = 10\,Hz \cdot 2\pi = 62.8$ *rad/s? Is it physically infeasible in some other way? If so, reject the model.*

write the answer here

Goodness of fit

Does the model give a low residual?

write the answer here

Problem 2 — Control Design

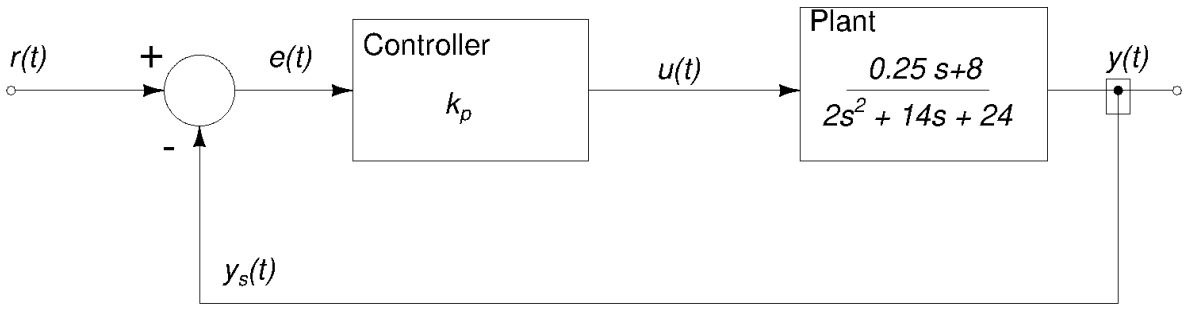


Figure 5: System with p-type control.

Test values of k_p on the range $k_p = 0.01 \ldots 100$. Select the value of k_p that best meets these design goals:

Measure	Goal
Rise time	Less than 1/2 second
Settling time	Less than 2 seconds
Steady-state error	As small as possible
Overshoot	Less than 10%

Steady-state error for a step input is defined as $e(t)$ after $y(t)$ reaches its steady value.

Problem 2.A

What is the suitable value of k_p?

write the answer here

Problem 2.B

As k_p increases, does rise time increase or decrease?

write the answer here

Problem 2.C

As k_p increases, does settling time increase or decrease?

write the answer here

Problem 2.D

As k_p increases, does steady-state error increase or decrease?

write the answer here

Problem 2.E

As k_p increases, does overshoot increase or decrease?

write the answer here